

Completed Topology

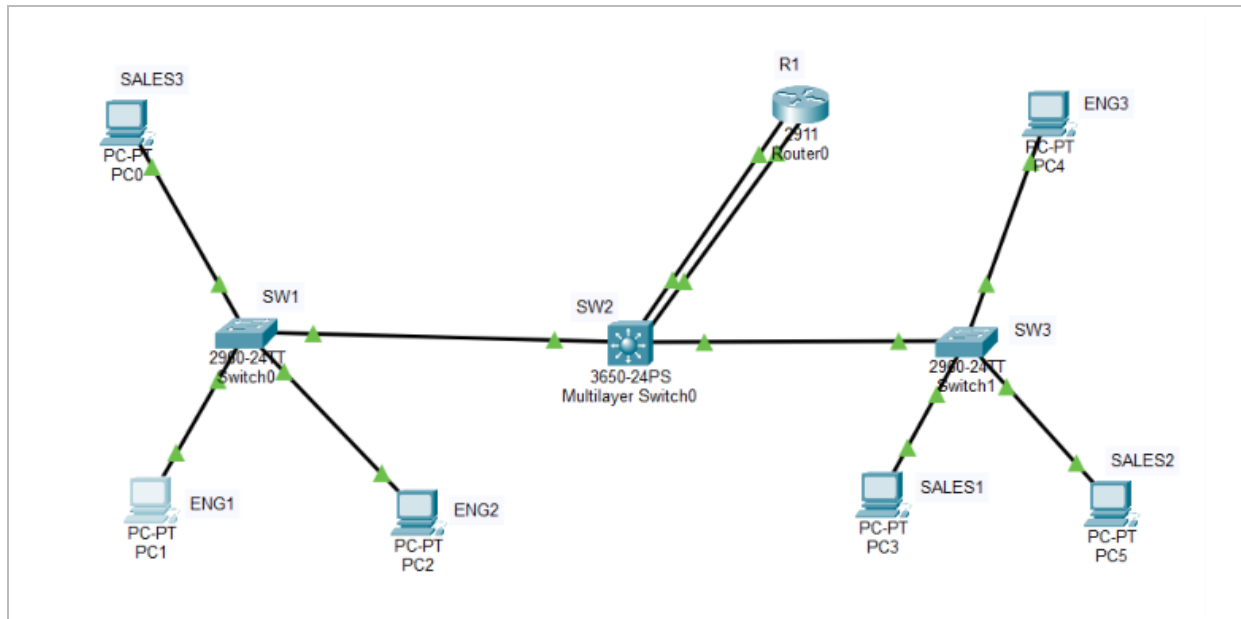


Figure 1: Completed topology in Packet Tracer

Task 89: VTP, Domain, Mode and Password

The VTP settings I chose are in the table below. SW2 is the server, and SW1 and SW3 are clients.

Setting	Value
VTP domain name	CAMPUS
VTP password	cisco123
SW2 (3650-24PS) mode	Server
SW1 and SW3 mode	Client

```

%SYS-5-CONFIG_I: Configured from console by console

SW2#conf t
Enter configuration commands, one per line. End with CNTL/
SW2(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
SW2(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name

SW2(config)#vtp password cisco123
Setting device VLAN database password to cisco123
SW2(config)#vtp mode server
Device mode already VTP SERVER.
SW2(config)#vlan 10
SW2(config-vlan)#name ENG
SW2(config-vlan)#vlan 20
SW2(config-vlan)#name SALES
SW2(config-vlan)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#

```

Figure 2: SW2: VTP domain, password and server mode set, then VLAN 10 (ENG) and VLAN 20 (SALES) created

```

SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z
SW1(config)#vtp domain CAMPUS
Domain name already set to CAMPUS.
SW1(config)#vtp password cisco123
Setting device VLAN database password to cisco123
SW1(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure 3: SW1: VTP domain, password and client mode set

```

SW2#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : CAMPUS
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.421B.6D00
Configuration last modified by 0.0.0.0 at 3-1-93 00:09:31
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 7
Configuration Revision   : 4
MD5 digest               : 0x27 0x94 0x7F 0xF4 0x29 0x4E 0xD1 0xA6
                        : 0xD0 0x8C 0xFF 0x6B 0xAE 0x8D 0x6F 0xCD
SW2#

```

Figure 4: SW2: show vtp status

```

SW3#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1

10   ENG                    active
20   SALES                  active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

SW3#show vtp status
VTP Version capable       : 1 to 2
VTP version running       : 1
VTP Domain Name           : CAMPUS
VTP Pruning Mode          : Disabled
VTP Traps Generation      : Disabled
Device ID                 : 000C.CF99.8300
Configuration last modified by 0.0.0.0 at 3-1-93 00:09:31

Feature VLAN :
-----
VTP Operating Mode        : Client
Maximum VLANs supported locally : 255
Number of existing VLANs  : 7
Configuration Revision     : 4
MD5 digest                : 0x27 0x94 0x7F 0xF4 0x29 0x4E 0xD1 0xA6
                           0xD0 0x8C 0xFF 0x6B 0xAE 0x8D 0x6F 0xCD

SW3#

```

Figure 5: SW3: show vtp status

```

SW1>en
SW1#show vtp status
VTP Version capable       : 1 to 2
VTP version running       : 1
VTP Domain Name           : CAMPUS
VTP Pruning Mode          : Disabled
VTP Traps Generation      : Disabled
Device ID                 : 0060.7051.EB00
Configuration last modified by 0.0.0.0 at 3-1-93 00:09:31

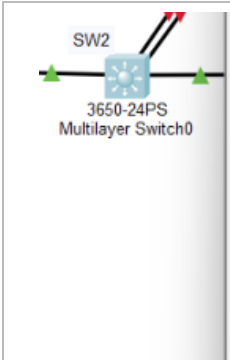
Feature VLAN :
-----
VTP Operating Mode        : Client
Maximum VLANs supported locally : 255
Number of existing VLANs  : 7
Configuration Revision     : 4
MD5 digest                : 0x27 0x94 0x7F 0xF4 0x29 0x4E 0xD1 0xA6
                           0xD0 0x8C 0xFF 0x6B 0xAE 0x8D 0x6F 0xCD

SW1#

```

Task 90: Create VLANs 10 and 20

VLAN 10 (ENG) and VLAN 20 (SALES) were created on SW2, the VTP server (see Figure 2 for the commands). SW2 lists both VLANs below. SW3 also shows them in Figure 5, which proves VTP is passing them on.



```

SW2#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6
                                           Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10
                                           Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14
                                           Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18
                                           Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22
                                           Gig1/0/23, Gig1/0/24, Gig1/1/1, Gig1/1/2
                                           Gig1/1/3, Gig1/1/4

10   ENG                    active
20   SALES                  active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

SW2#

```

Figure 6: SW2: show vlan brief showing VLAN 10 ENG and VLAN 20 SALES

Task 91: Trunk Ports and Access Ports

Trunk ports carry both VLANs between the switches. Access ports connect each PC to the correct VLAN: ENG hosts to VLAN 10 and SALES hosts to VLAN 20.

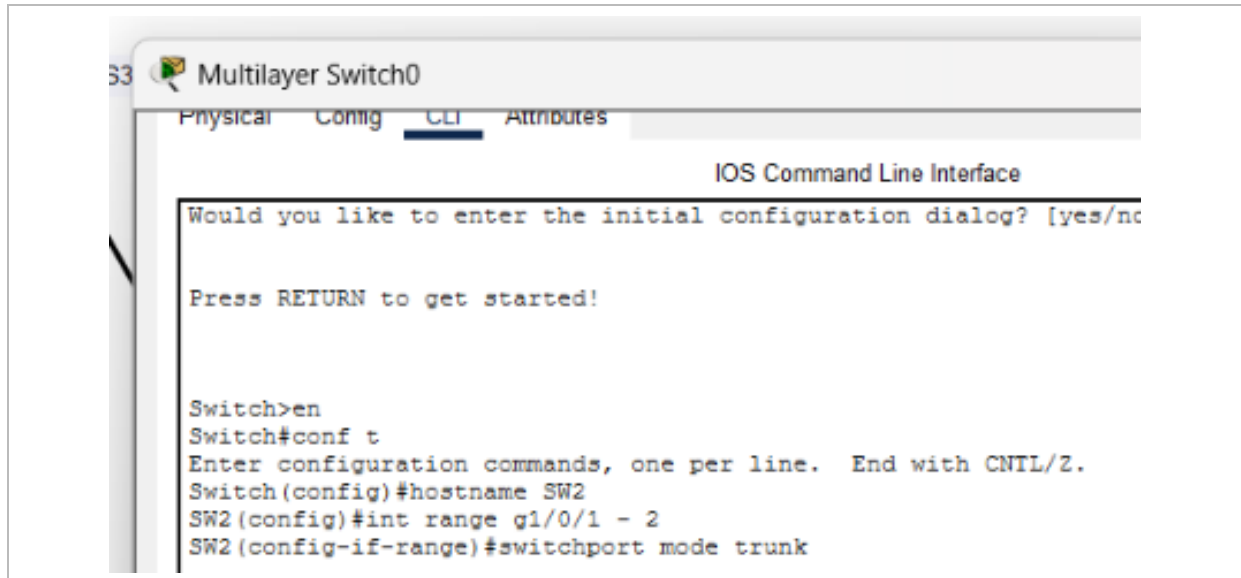


Figure 7: SW2: trunk mode set on Gi1/0/1 and Gi1/0/2 (the links to SW1 and SW3)

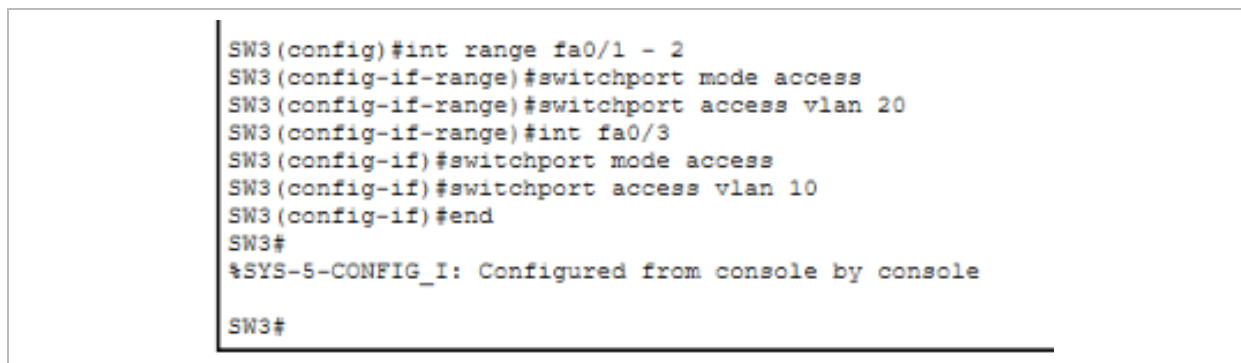


Figure 8: SW3: access ports set (Fa0/1 and Fa0/2 in VLAN 20 for SALES1 and SALES2, Fa0/3 in VLAN 10 for ENG3)

Task 92: Verify VLAN and Trunk Configuration

The commands show vlan brief and show interfaces trunk were run to check the setup. On SW3, Gig0/2 is trunking and the PCs are in the right VLANs: Fa0/3 (ENG3) in VLAN 10, and Fa0/1 and Fa0/2 (SALES1, SALES2) in VLAN 20.

```
SW3#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/2    auto      n-802.1q       trunking      1

Port      Vlans allowed on trunk
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/2    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/2    1,10,20

SW3#show vlan brief
VLAN Name                Status    Ports
-----
1    default              active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1
10   ENG                  active    Fa0/3
20   SALES                active    Fa0/1, Fa0/2
1002 fddi-default       active
1003 token-ring-default active
1004 fddinet-default    active
1005 trnet-default      active
SW3#
```

Figure 9: SW3: show interfaces trunk and show vlan brief

Task 93: Inter-VLAN Routing Option 1 (Separate Interfaces on the Router)

R1 uses one interface per VLAN. Gi0/0 connects to SW2 in VLAN 10 and has the address 10.10.10.1. Gi0/1 connects to SW2 in VLAN 20 and has the address 10.10.20.1. Both addresses are the default gateways for their VLANs.

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int g0/1
R1(config-if)#ip address 10.10.20.1 255.255.255.0
R1(config-if)#no shutdown
```

Figure 10: R1: IP addresses set on Gi0/0 (10.10.10.1) and Gi0/1 (10.10.20.1)

```

R1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.10.1      YES manual up          up
GigabitEthernet0/1  10.10.20.1      YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
R1#

```

Figure 11: R1: show ip int brief (both interfaces up/up)

```

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time=1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

Figure 12: Ping from ENG1 (VLAN 10) to SALES1 (10.10.20.10): 4 sent, 4 received, 0% loss

Task 94: Inter-VLAN Routing Option 2 (Router on a Stick)

Only one link is used. R1 Gi0/0 is split into two subinterfaces: Gi0/0.10 for VLAN 10 and Gi0/0.20 for VLAN 20, each using 802.1Q. The second link (Gi0/1) is shut down and unused.

```

Vlan1           unassigned      YES unset  administratively down down
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/1
R1(config-if)#no ip address
R1(config-if)#shutdown

R1(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

R1(config-if)#int g0/0
R1(config-if)#no ip address
R1(config-if)#no shutdown
R1(config-if)#int g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to down

R1(config-subif)#encapsulation dotQ 10
^
% Invalid input detected at '^' marker.

R1(config-subif)#encapsulation dot1q
^
% Incomplete command.
R1(config-subif)#encapsulation dot1q 10
R1(config-subif)#ip address 10.10.10.1 255.255.255.0
R1(config-subif)#int g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to down

R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip address 10.10.20.1 255.255.255.0
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#show ip int brief
^
% Invalid input detected at '^' marker.

R1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES manual up          up
GigabitEthernet0/0.10  10.10.10.1      YES manual up          up
GigabitEthernet0/0.20  10.10.20.1      YES manual up          up
GigabitEthernet0/1  unassigned      YES manual administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
R1#

```

Figure 13: R1: Gi0/1 shut down, subinterfaces Gi0/0.10 and Gi0/0.20 created with dot1q encapsulation

```

R1#show ip int brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES manual up          up
GigabitEthernet0/0.10 10.10.10.1     YES manual up          up
GigabitEthernet0/0.20 10.10.20.1     YES manual up          up
GigabitEthernet0/1    unassigned      YES manual administratively down down
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Vlan1                unassigned      YES unset  administratively down down
R1#

```

Figure 14: R1: show ip int brief (subinterfaces .10 and .20 are up/up, Gi0/1 is administratively down)

```

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 10.10.20.10

Tracing route to 10.10.20.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.10.10.1
  1  0 ms    0 ms    1 ms    10.10.20.10

Trace complete.

```

Figure 15: Ping and tracert from ENG1 to SALES1

Task 95: Inter-VLAN Routing Option 3 (Layer 3 Switch)

SW2 now routes between the VLANs itself. Routing was switched on with ip routing, and VLAN interfaces (SVIs) were created: Vlan10 with 10.10.10.1 and Vlan20 with 10.10.20.1. The router is not used.

```

SW2#show ip int brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet1/0/1 unassigned      YES unset up          up
GigabitEthernet1/0/2 unassigned      YES unset up          up
GigabitEthernet1/0/3 unassigned      YES unset administratively down down
GigabitEthernet1/0/4 unassigned      YES unset administratively down down
GigabitEthernet1/0/5 unassigned      YES unset down        down
GigabitEthernet1/0/6 unassigned      YES unset down        down
GigabitEthernet1/0/7 unassigned      YES unset down        down
GigabitEthernet1/0/8 unassigned      YES unset down        down
GigabitEthernet1/0/9 unassigned      YES unset down        down
GigabitEthernet1/0/10 unassigned      YES unset down        down

```

Figure 16: SW2: show ip int brief (Gi1/0/3 and Gi1/0/4 to the router are shut down)

```

SW2#
SW2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.10.0/24 is directly connected, Vlan10
L       10.10.10.1/32 is directly connected, Vlan10
C       10.10.20.0/24 is directly connected, Vlan20
L       10.10.20.1/32 is directly connected, Vlan20

SW2#

```

Figure 17: SW2: show ip route (VLAN 10 and VLAN 20 subnets are directly connected)

```

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time=5ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time=18ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 5ms

C:\>tracert 10.10.20.10

Tracing route to 10.10.20.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.10.10.1
  1  0 ms    0 ms    0 ms    10.10.20.10

Trace complete.

C:\>

```

Figure 18: Ping and tracert from ENG1 to SALES1 using SW2 as the Layer 3 switch

Task 96: Reflection Question

Option 3, the Layer 3 switch, would scale best. The switch does the routing inside itself, so traffic does not have to travel to a separate router and back. Option 1 needs one router port and one cable for every VLAN, so a large campus would quickly run out of ports. Option 2 pushes all VLANs through a single link, which becomes slow when traffic grows and is a single point of failure. With a Layer 3 switch, adding a new VLAN only needs a new VLAN and a new SVI, with no extra cabling.